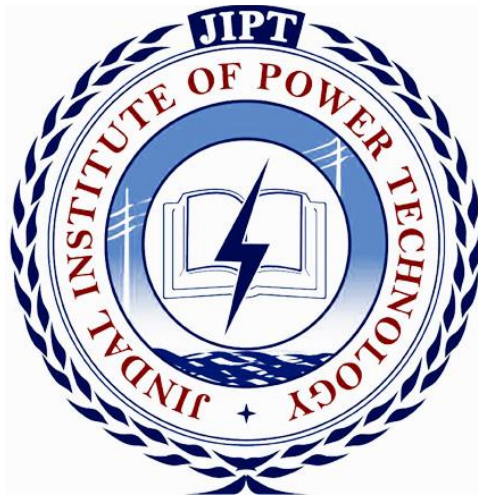


A
PROJECT REPORT
ON
TRIPPING AND THEIR ANALYSIS OF THERMAL POWER PLANTS



Submitted by:

SURAJ KUMAR (16PG047)

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Guided by:

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ASSISTANT PROFESSOR

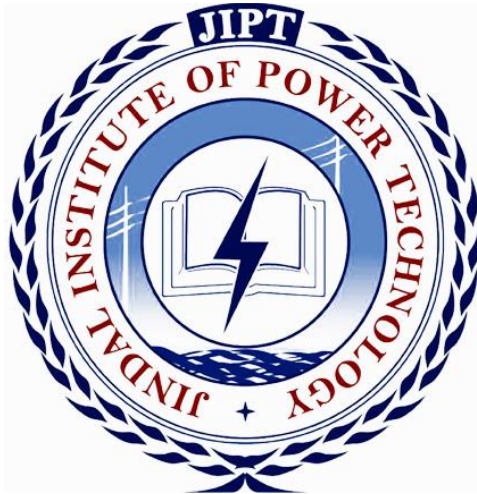
JIPT, TAMNAR, RAIGARH

JINDAL INSTITUTE OF POWER TECHNOLOGY (JIPT)

O. P. Jindal Super Thermal Power Plant

Vill & PO: Tamnar, Dist.: Raigarh Pin-496 107 (CG)

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ABSTRACT

A power station is subjected to various internal and external demanding parameters, which try to affect its reliability. A power-generating unit can trip due to various reasons. The main need for tripping is the safety of the human beings and the equipment's / sensors that sense these parameters may malfunction, and it also can lead to tripping of the unit. Each forced outage causes generation loss and hence revenue loss for the company. One tripping of a unit causes a loss of availability.

The well-run plants have a trip committee which is entrusted with the task of root cause analysis of trips and suggesting corrective actions to prevent recurrence of trips. The suggested corrective actions are typically formulated as an action plan with clearly earmarked responsibility center and schedule. Compliance with such recommendations is monitored at plant as well as corporate levels. Recommendations of the trip committee also feed into the maintenance plan.

In some cases, specialized committees are also in place for analyzing boiler tube leakages – one of the most frequent reasons for forced outages. Other causes of forced outage are also analyzed in detail by what are called as 'Forced Outage Committees'.

ACKNOWLEDGEMENT

We would like to articulate our deep gratitude to Jindal Institute of Power Technology, Tamnar for giving us this opportunity to carry out the project inside their premises. We express our heartiest thanks to Mr. S.K. Pal for giving us guidance and support during the course of our complete project.

We wish to express our profound sense of gratitude to Director & all the faculty members of Jindal Institute of Power Technology, Tamnar, Raigarh(C.H) for their delightful guidance and constant encouragement throughout the process; they have always been a great inspirational motivator for us.

We take this opportunity to express our whole hearted thanks & gratitude to other executives of JIPT & JPL for their involvement in the process, which make it possible to achieve the completion of this project in time.

SURAJ KUMAR

VIVEK MANDAL

MD REFATULLAH SK

SAURABH KUMAR JHA

Date:.....

Jindal Institute of Power Technology
Tamnar, Raigarh (CG)

CERTIFICATE

This is to certify that Mr. SURAJ KUMAR, Mr. VIVEK MANDAL, Mr. MD REFATULLAH SK and Mr. SAURABH KUMAR JHA have submitted the project entitled “Tripping and its Analysis for Thermal Power Plants” as a partial requirement for completion of Post Graduate Program in Thermal Power Plant Technology. This work has been carried out by them under my guidance and supervision.

For completion of this project they have worked with determination & dedication. I wish them best of luck and a very bright future.

S K Pal

Assistant Professor

Jindal Institute of Power Technology
Tamnar, Raigarh (C.H.)

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Chapter 1

INTRODUCTION

1.1 AIM& OBJECTIVES:

The main aim of this project is to discuss about the past tripping happen in the power plant and also to discuss the symptoms of the tripping with main cause of tripping and the action taken to prevent further tripping of the same component with the recommendation given to prevent further tripping of the same auxiliary.

There are different types of tripping happen in the power plant some have their causes and some are human error. To identify these type of tripping is also the main aim of these project.

To learn from the mistakes of other is the Human Characteristic same apply for the Power Plant to learn from the tripping of other power plant and to prevent there system from tripping so to save lots of available power and to increase the life of the power plant.

1.2 SCOPE & LIMITATIONS:

It is difficult to accurately diagnose and locate faults in the thermal system of power plants due to the coupling of multiple components.

There are different types of tripping happen in the power plant, some of which have their causes pertaining to inherent defects and others are human error. Some causes of tripping are easily analyzed and some are hard to find the cause of the malfunctioning. Study focused on available resource material at JIPT & JPL inclusive of previous experiences and exposure of different authorities.

1.3 METHODOLOGY:

For completing this project, we have used two methodologies. One is directly getting required document from the JIPT library and other is contacting nearby plant to get the desired document from there monitoring department. Plants like JPL, BHUSHAN STEEL & POWER, NTPC and TATA POWER are some of the plant, which helped us to complete our project on time.

The some of the data used in this project report are gathered from the JIPT Library and the mentor desk.

We have utilized a library consisting of all the materials we need to do our project work. Our mentor also help us a lot in order to get materials required for the case study and to complete following project work.

1.4 FUTURE LINKAGES:

To learn from the mistakes of other is the Human Characteristic same apply for the Power Plant to learn from the tripping of other power plant and to prevent there system from tripping so to save lots of available power and to increase the life of the power plant.

This project help the new power plant to learn and to prevent the tripping of their auxiliaries and to prevent loss of available power.

1.5 CONCLUSION:

Now days tripping affect the most of the power plant availability factor. Due to this tripping a power plant faces many problems such as:

- Loss of auxiliary
- Loss of availability of the plant.
- Loss of available power.
- Loss in profit of the plant.

This study help us to understand the main causes behind the tripping of any auxiliary and also unable any to prevent the same tripping happen again in future.

Chapter 2

LITERATURE SURVEY

2.1 Introduction

A power station is subjected to various internal and external demanding parameters, which try to affect its reliability. A power-generating unit can trip due to various reasons. The main need for tripping is the safety of the human beings and the equipment's / sensors that sense these parameters may malfunction, and it also can lead to tripping of the unit. Each forced outage causes generation loss and hence revenue loss for the company. One tripping of a unit causes a loss of availability.

2.2 Tripping Analysis

To analyse each and every tripping and identify the causes, a "Tripping Analysis Team" is established. The basic function of the "Tripping Analysis Team" was to analyse and arrive at the exact cause of the tripping, and also to recommend any modification in the system. The modifications suggested by the "tripping analysis team" are accepted by the Original Equipment Manufacturer and implemented. The success of this system can be seen by the difference in number of tripping of two units in initial days.

2.3 Event Analysis

To further strengthen the tripping analysis team, a system of "Event analysis" can be established. Here the difference is that any event, which has potential of leading towards tripping or even generation loss, is identified and analysed. This helps in avoiding generation loss as well as in many cases tripping of unit. This analysis also gives suggestions for system and procedures modifications.

These reports are reviewed at various levels up to divisional head and circulated for disseminations.

2.4 Conclusion

In order to build-upon the shared expertise across various power plants, a Pool of Technical Experts is developed across the organization, deriving expertise in different areas (such as turbine, boilers, C&I etc.) from different locations. From this pool, Knowledge Teams are derived, which bring knowledge and experience in different areas from different power plants and provide in-house consultancy to technical problems at any location.

Chapter 3

TRIPING CASE STUDIES

Conducting Regular Seminar on Thermal Power Plant Problems faced and its solutions done will help to reduce the hazards in Power Plants.

Case Study – 01.

Name of the Power Plant: BAKRESHWAR THERMAL POWER STATION, CHINPAI AND BHURKUNA GRAM PANCHAYAT AREAS OF BIRBHUM DISTRICT, WEST BENGAL

Unit Running Status:

Load: 250MW

Date: 04.09.08

Name of the Equipment Failed: FD FAN-A/B BLADE PITCH MIN.

Cause: Lube oil pressure decrease due to filter choking, control oil pr. also decrease.

Observations:

- Oil Pressure after filter started dropping
- Lub oil pressure hi alarm appeared.
- FD fan blade pitch went to minimum.
- ID fan speed decreased to maintain Furnace Pressure.
- FD Fan bearing temperature increased and Br. Temp. hi alarm appeared at 90 °C
- Fuel Flow reduced.
- The feed flow and steam flow decreased.
- Machine Load decreased.
- FD Fan got tripped through bearing Temperature very hi.

Root Cause of failure:

- ❖ Lube oil pressure decrease due to filter choking.
- ❖ Control oil pr. also decrease.

Action Taken:

- ✓ Filter changed.

Recommendations:

- Dp across Lub oil strainer is to be checked regularly and if Dp is seen increasing Lub oil strainer is to be changed over and clean the choked strainer.

Case Study – 02

Name of the Power Plant: BAKRESHWAR THERMAL POWER STATION, CHINPAI AND BHURKUNA GRAM PANCHAYAT AREAS OF BIRBHUM DISTRICT, WEST BENGAL

Unit Running Status:

Load: 250MW

Date: 06.01.08

Name of the Equipment Failed: LOW FURNACE PRESSURE.

Cause: DCS output signal failure.

Observations:

- Activation of this malfunction causes the DCS furnace pressure output signal to go high.
- IDF-A and IDF-B speed decrease, attempting to control the furnace pressure.
- ID fan inlet damper position decrease in order to additionally control the furnace pressure.
- IDF-A and IDF-B motor current decrease.
- Furnace pressure increases. Furnace high pressure alarm appeared.
- FD fan blade pitch goes to minimum.

Root Cause of failure:

- ❖ DCS output signal failure.

Note: Operator does not took any action in order to transfer individual IDF controller to manual control, it result in boiler master fuel trip on high furnace pressure. Removal of this malfunction return the faulty DCS circuit to normal.

Action Taken:

- ✓ DCS scheme was modified.

Recommendations:

- Regular checking DCS for any fault.
- Provision for monthly DCS maintenance.

Case Study – 03.

Name of the Power Plant: BAKRESHWAR THERMAL POWER STATION, CHINPAI AND BHURKUNA GRAM PANCHAYAT AREAS OF BIRBHUM DISTRICT, WEST BENGAL

Unit Running Status:

Load: 250MW

Date: 04.10.10

Name of the Equipment Failed: SYSTEM FREQUENCY LOW.

Cause: High load demand but low generation.

Observations:

- Activation of this malfunction at 100% load drop the system frequency low due to outside grid high load demand.
- Turbine speed decreases.
- Low frequency alarm comes in.
- Turbine speed controller act to maintain the turbine speed, increasing the turbine control valves position. Throttle pressure decreases.
- FD fan and ID fan speed decrease due to frequency decrease, causing the flue and air flow to decrease.
- Main turbine trip on low frequency.
- Removal of this malfunction return the outside grid load to normal.

Root Cause of failure:

- ❖ High load demand but low generation.

Action Taken:

- ✓ Grid frequency monitored and accordingly load was changed.

Recommendations:

- Continuous monitoring of Grid Frequency.

Case Study – 04.

Name of the Power Plant: BAKRESHWAR THERMAL POWER STATION, CHINPAI AND BHURKUNA GRAM PANCHAYAT AREAS OF BIRBHUM DISTRICT, WEST BENGAL

Unit Running Status:

Load: 250MW

Date: 07.07.08

Name of the Equipment Failed: COAL HANGUP IN MILL.

Cause: Wet coal resulting in chute blockage.

Observations:

- Activation of this malfunction cause the coal to hang up in mill.
- Mill A outlet coal air mixture temperature decreases. Mill A hot air damper goes wide open and cold air damper closes to minimum position in an attempt to control the temperature.
- Mill A motor current increases. Total coal rate decreases.
- Mill A bowl high differential pressure alarm comes on. Mill A primary air flow decreases.
- All remaining coal feeders feed rate and total coal feed rate increase then stabilize at a higher rate.
- Superheater steam pressure decreases, then restores to normal.

Root Cause of failure:

- ❖ Wet coal resulting in chute blockage.

Action Taken:

- ✓ Bunker was empty and fresh coal was feed.
- ✓ Wet coal was removed from Feeder and mill.

Recommendations:

- Regular checking of coal quality to be carried out before feeding.
- Provision for coal quality check in the bunker.

Case Study – 05.

Name of the Power Plant: BAKRESHWAR THERMAL POWER STATION, CHINPAI AND BHURKUNA GRAM PANCHAYAT AREAS OF BIRBHUM DISTRICT, WEST BENGAL

Unit Running Status:

Load: 250MW

Date: 04.09.05

Name of the Equipment Failed: LUBE OIL SUPPLY FAILURE TO PA FANS.

Cause: Strainers clogging.

Observations:

- Activation of this malfunction cause the PA A fan lube oil filter A to clog.

- Delta p across the filters increases under the condition that the oil filter A was in operation.
- PA A fan lube oil pressure drop. Low oil pressure alarm activated and PA A fan trip on bearing temperature high.
- PA A fan inlet flow and discharge pressure drop. Mill A and feeder A trip. Total coal feed rate decreases.
- Superheater steam pressure decreases.
- Generator output power decreases.
- Remaining coal feeders feed rate increases then stabilize at a higher rate.
- Superheater steam pressure increase, then stabilize at normal value. Generator output power increase, then stabilize.

Root Cause of failure:

- ❖ Strainers clogging.
- ❖ No Alarm regarding differential pressure.
- ❖ **Human Error.**

Action Taken:

- ✓ Strainers Changed and at the place of one two strainers place.

Recommendations:

- Regular checking of strainer's health to be carried out.

Case Study – 06.

Name of the Power Plant: BAKRESHWAR THERMAL POWER STATION, CHINPAI AND BHURKUNA GRAM PANCHAYAT AREAS OF BIRBHUM DISTRICT, WEST BENGAL

Unit Running Status:

Load: 250MW

Date: 23.06.09

Name of the Equipment Failed: WINDBOX TO FURNACE DIFFERENTIAL PRESSURE HIGH/LOW.

Cause: Secondary air damper fails to modulate.

Observations:

- Activation of this malfunction cause the secondary air damper faulty modulation.
- Windbox to furnace differential pressure and furnace pressure start going up and down.

- Secondary air flow starts fluctuating. Fuel/air flow difference fluctuate.
- FD fan blade pitch change accordingly in order to maintain the total air flow.
- Removal of this malfunction return the faulty air damper to normal.
- ID fans scoop and IGV change to maintain furnace pr.

Root Cause of failure:

- ❖ Secondary air damper fails to modulate.

Action Taken:

- ✓ Secondary air damper check for fault.

Recommendations:

- Regular checking of Air dampers to be carried out with some time interval.
- Provision for maintenance of this dampers.

Case Study – 07.

Name of the Power Plant: BAKRESHWAR THERMAL POWER STATION, CHINPAI AND BHURKUNA GRAM PANCHAYAT AREAS OF BIRBHUM DISTRICT, WEST BENGAL

Unit Running Status:

Load: 250MW

Date: 10.09.12

Name of the Equipment Failed: DEAERATOR LEVEL HIGH.

Cause: Level sensor fails high (Control Transmitter).

Observations:

- A low level is annunciated and then the very low level trip setpoint is reached. Deaerator level control can be restored by placing the Deaerator Level Controller in manual. Malfunction removal restores normal operation of the level transmitter.
- The Deaerator Level Controller sends a close signal to the DM Makeup to Condenser valve and an open signal to the Condenser Excess Return valve. Actual deaerator level decreases.
- Activation of this malfunction fails one of two Deaerator Tank level transmitter outputs to the high end of the range, but it was low enough that the DCS averages it with the other level transmitter rather than ignoring the signal.

Root Cause of failure:

- ❖ Level sensor fails high (Control Transmitter).
- ❖ PLCC panel malfunctioning.

Action Taken:

- ✓ Change of the transmitter and PLCC circuit.

Recommendations:

- Regular checking of level controller and PLCC panel to be carried out.
- Provision for regular level check.

Case Study – 08.

Name of the Power Plant: BAKRESHWAR THERMAL POWER STATION, CHINPAI AND BHURKUNA GRAM PANCHAYAT AREAS OF BIRBHUM DISTRICT, WEST BENGAL

Unit Running Status:

Load: 250MW

Date: 15.05.08

Name of the Equipment Failed: TURBINE AUXILIARY STEAM PRESSURE LOW.

Cause: Sudden closing of control valve due to DCS malfunction.

Observations:

- Activation of this malfunction fails the DCS signal to Turbine Aux Steam Header High Capacity Press Control valve, and the valve goes fully shut. The valve cannot be reopened while the malfunction was active.
- If high capacity valve was supplying the Turbine Aux Steam system, header pressure decrease and a low pressure alarm activated.
- Steam pressure available for Turbine Gland Sealing, Deaerator Pegging and Air Ejector operation decrease.
- Turbine Aux Steam header pressure was restored by bypassing AS-22 with bypass valve.
- Malfunction removal restores the proper DCS control signal and normal operation of the valve.

Root Cause of failure:

- ❖ Sudden closing of control valve due to DCS malfunction.
- ❖ Due to this, no breaker could be operated.
- ❖ **In this unit, both DC supplies fed from a same Bus**

Action Taken:

- ✓ Scheme modified to ensure that each DC supply comes from different bus.

Recommendations:

- Regular checking of availability of DCS to be carried out.

Case Study – 09.

Name of the Power Plant: BAKRESHWAR THERMAL POWER STATION, CHINPAI AND BHURKUNA GRAM PANCHAYAT AREAS OF BIRBHUM DISTRICT, WEST BENGAL

Unit Running Status:

Load: 250MW

Date: 04.09.07

Name of the Equipment Failed: GLAND STEAM TEMPERATURE HIGH (pr. abnormal).

Cause: Inlet control valve frozen.

Observations:

- Activation of this malfunction binds the Gland Steam Pressure Control valve, at its current position. The valve cannot be moved while the malfunction was active.
- If the malfunction was activated at low turbine load, increasing turbine load increases the supply of steam from the HP turbine to the gland seal header.
- Gland seal header pressure increases with no further closing of the supply valve.
- If the gland seal header steam supply from gland seal controller and turbine leak off were within the capacity of the gland seal dump valve, Gland Seal pressure was controlled.
- If the supply exceeds the dump capacity, pressure (and temperature) continue to increase resulting in GS system relief valve operation.

Root Cause of failure:

- ❖ Inlet control valve frozen.

Action Taken:

- ✓ Inlet control valve changes.

Recommendations:

- Regular checking of inlet control valve to be carried out.
- Gland seal header pressure to be monitored continuously.

Case Study – 10

Name of the Power Plant: BAKRESHWAR THERMAL POWER STATION, CHINPAI AND BHURKUNA GRAM PANCHAYAT AREAS OF BIRBHUM DISTRICT, WEST BENGAL

Unit Running Status:

Load: 250MW

Date: 24.05.08

Name of the Equipment Failed: HP BYPASS WARMUP VALVE (LEFT) FAIL OPEN.

Cause: Spurious signal.

Observations:

- Activation of this malfunction fails the HP Bypass Valve Warmup Line isolation valve, full open.
- The valve not respond to any close command.
- During normal plant operation, some steam flow from the main steam header bypass the HP turbine through the warmup line.
- If the HP Bypass Station was in operation, some flow bypass the HP Bypass Desuperheater.
- Malfunction removal restores normal operation of the valve.

Root Cause of failure:

- ❖ Spurious signal.
- ❖ Malfunctioning of PLCC Panel.

Action Taken:

- ✓ Circuit change in PLCC panel.

Recommendations:

- Regular checking of PLCC panel to be carried out.
- Provision for maintenance of PLCC panel.

Case Study – 11.

Name of the Power Plant: BAKRESHWAR THERMAL POWER STATION, CHINPAI AND BHURKUNA GRAM PANCHAYAT AREAS OF BIRBHUM DISTRICT, WEST BENGAL

Unit Running Status:

Load: 250MW

Date: 06.09.12

Name of the Equipment Failed: GENERATOR SEAL OIL TEMPERATURE HIGH.

Cause: Seal oil cooler gets fouled (due to deposits).

Observations:

- Activation of this malfunction reduces the heat transfer coefficient in the affected generator seal oil cooler, resulting in a reduced heat transfer between the oil and the cooling water.
- The seal oil temperature leaving the cooler increases.
- Seal oil system oil-side temperatures continuously increases.
- Normal seal oil cooling restored by switching to the unaffected seal oil cooler.
- Malfunction removal restores normal heat.

Root Cause of failure:

- ❖ Seal oil cooler gets fouled (due to deposits).
- ❖ Continues unwanted material cause deposit.
- ❖ **Generator metal getting removed.**

Action Taken:

- ✓ Cooler changed.
- ✓ Generator metal checked and changed.

Recommendations:

- Regular checking of filter and cooler condition to be carried out.
- Provision for extra filter to be provided on site.

Case Study – 12.

Name of the Power Plant: FARAKKA SUPER THERMAL POWER STATION, NTPC COAL STATION, FARAKKA (W.B)

Unit Running Status:

Load: 200MW

Date: 24.06.09

Name of the Equipment Failed: FLUCTUATION IN BFP DISCHARGE PRESSURE.

Cause: Due to failure of Card.

Observations:

- BFP discharge header pressure indication went bad, drum level came down.
- Feed Master demand of BFPs increased to its maximum. But speed control demand of TDBFPs did not change and drum level went low.

- Mill D was tripped and MDBFP was started. Drum level increased but could not be controlled as MDBFP scoop (in auto) was in maximum.
- Boiler tripped in drum level hi-hi protection.
- Due to failure of Card, Speed control of both TDBFPs got shifted to manual (Status change indication is not available in Operating panel or alarm facia). Hence it did not respond to the feed master demand.

Root Cause of failure:

- ❖ Due to failure of Card, Speed control of both TDBFPs got shifted to manual.
- ❖ Speed control demand of TDBFPs did not change and drum level changes.

Action Taken:

- ✓ Card replaced with new one.
- ✓ Program restored.

Recommendations:

- Regular checking of Card to be carried out.
- Provision for extra card to be provided.

Case Study – 13.

Name of the Power Plant: FARAKKA SUPER THERMAL POWER STATION, NTPC COAL STATION, FARAKKA (W.B)

Unit Running Status:

Load: 250MW

Date: 04.03.10

Name of the Equipment Failed: BUS BAR FAILURE DUE TO POOR CABLING WORK.

Cause: Cabling work.

Observations:

- Major parameters indications became bad. ID Fan A&B IGV became inoperative.
- Furnace pressure went up in spite of cutting out mill A&B.
- Super heater spray controller also got inoperative from OWS and Main steam temp. started roping.
- As controls were not working, Boiler was hand tripped.
- DDCMIS 5Volt power supply in Node-1 second last and last rack was found lower than required.

- It was observed that the cable layout was causing stress on the bus bar and intermittently causing faulty connection.

Root Cause of failure:

- ❖ Cable layout was causing stress on the bus bar.
- ❖ Intermittently causing faulty connection.

Action Taken:

- ✓ Scheme modified to ensure no poor cable layout for the bus bar.
- ✓ Realignment of bus bar.

Recommendations:

- All cables layout should check and altered if any.

Case Study –14.

Name of the Power Plant: FARAKKA SUPER THERMAL POWER STATION, NTPC COAL STATION, FARAKKA (W.B)

Unit Running Status:

Load: 170MW

Date: 08.03.10

Name of the Equipment Failed: FSSS ACTIVATED.

Cause: Failure of RET memory card.

Observations:

- Unit was running at 170MW. All of a sudden both PA fans tripped and Boiler tripped on “Flame failure” protection. DDCMIS input 19 signal of MFR-B became high from FSSS and got reset within 2secs.
- Failure of RET memory card of FSSS caused status high of MFR B.
- Due to this faulty signal generation, Tripping of both PA Fans initiated from this signal, though MFR-B in FSSS did not really actuate.

Root Cause of failure:

- ❖ Failure of RET memory card cause “FLAME FAILURE” signal.

Action Taken:

- ✓ RET memory card changed.

Recommendations:

- Regular checking for RET card to be carried out.

Case Study – 15.

Name of the Power Plant: FARAKKA SUPER THERMAL POWER STATION, NTPC COAL STATION, FARAKKA (W.B)

Unit Running Status:

Load: 250MW

Date: 11.03.10

Name of the Equipment Failed: BOILER TRIPPED ON “FURNACE PRESSURE LO-LO”.

Cause: PA outlet damper malfunction.

Observations:

- “PA Fan A Approaching stall” & “PA Fan A suction/ discharge DP Low” alarm appeared. Furnace pressure went low and subsequently PA Fan A tripped.
- Boiler tripped on “Furnace pressure lo-lo” protection. It was observed that “PA Fan A approaching stall”; alarm appeared and after 2 seconds, “PA Fan A outlet damper close”, “PA Fan A Trip” and “PA Fan A outlet damper malfunction” had appeared simultaneously. No flag appeared at PA Fan A breaker.
- As per trip logic, PA Fan will trip if two out of two “discharge damper not open” limit switches are actuated. As PA Fan – A "suction / discharge DP low" and "PA Fan - A approaching stall" alarm appeared along with PA Fan tripping, it was concluded that PA Fan - A tripped due to actual closing of discharge damper.
- Closure of discharge damper has been defeated by isolation of air supply. Monitoring of initiation of discharge damper close command provided through chartless recorder and is under observation.

Root Cause of failure:

- ❖ PA outlet damper malfunction.

Action Taken:

- ✓ Closure of discharge damper has been defeated by isolation of air supply.
- ✓ Monitoring of initiation of discharge damper close command provided through chartless recorder and is under observation.

Recommendations:

- Regular checking of command of discharge damper closing to be carried out.

Case Study – 16.

Name of the Power Plant: TANDA THERMAL POWER STATION, AMBEDKAR NAGAR, UTTAR PRADESH.

Unit Running Status:

Load: 113 MW

Date: 15.06.09

Name of the Equipment Failed: UNIT TRIP.

Cause: HRH temperature defective reading.

Observations:

- Unit was running at 113 MW with mills ABCDEF.
- Turbine tripped on HRH temperature very low protection, although HRH temperature was normal.
- The value for monitoring HRH temperature is taken from chartless recorder for further processing in the 2/3 logic formation for protection.
- The recorder got defective leading to tripping of the unit.

Root Cause of failure:

- ❖ HRH temperature very low.
- ❖ False reading.
- ❖ Temperature recorder got defective leading to tripping of the unit.

Action Taken:

- ✓ Recorder replaced.
- ✓ Logic reformatted.

Recommendations:

- Regular checking of proper working of recorder to be carried out.

Case Study – 17.

Name of the Power Plant: TANDA THERMAL POWER STATION, AMBEDKAR NAGAR, UTTAR PRADESH.

Unit Running Status:

Load: 120MW

Date: 23.12.09

- **Name of the Equipment Failed:** UNIT TRIPPED ON “FURNACE DRAFT LOW”.

Cause: Due to momentary break in power supply of DPUs.

Observations:

- Feeder's commands got reduced to zero along with reduction in PA fans loading.
- Unit tripped on "Furnace draft Low".
- The parameters on the workstation observed frozen.
- Unit tripped within two minutes of parameters freezing. It was found that many parameters originating from Old MAX CLCS panel had logged as "bad data" in the historian.
- The system power supply is taken from 23 separate PSUs. Any dip in the PSU O/P voltage, causes the DPUs to reinitialize.
- In the process of re-initialization, the DPUs fetch data from historian. Due to momentary break in power supply of DPUs, the historian receives bad data and on restoration of power supply the DPUs get re-initialized with the 'bad data' causing erratic action.

Root Cause of failure:

- ❖ Due to momentary break in power supply of DPUs.
- ❖ Due to bad data.

Action Taken:

- ✓ Scheme modified to ensure no break in power supply to DPU's.

Recommendations:

- DPU's panel to replace by new panel.
- DPU's to be connected with battery power

Case Study – 18.

Name of the Power Plant: KORBA SUPER THERMAL POWER PLANT, KORBA,
CHHATISGARH

Unit Running Status:

Load: 250MW

Date: 27.08.09

Name of the Equipment Failed: UNIT TRIPPED ON TURBINE TRIP AND DRUM LEVEL
LOW PROTECTION.

Cause: trip oil pressure low.

Observations:

- ‘Turbine tripped’ alarm appeared even though turbine was running.
- All extraction valves got closed. TD BFP A tripped on over speed during steam supply change over. MD BFP started in auto but tripped on hydro coupling bearing temperature protection.
- Unit tripped on drum level low protection.
- During last overhaul trip oil pressure low protection logic was changed from 1 out of 2 to 2 out of 3 logic.
- The above modification was carried out using Instamatic B card VU-11, VO-11 & FA-12 cards. It is suspected that modification of VU-11, VO-11 and associated circuit caused spurious initiation of trip oil pressure low signal. The protection has been restored to original scheme.
- The above cards kept under continuous monitoring.

Root Cause of failure:

- ❖ Unit tripped on turbine trip and drum level low protection.
- ❖ MD BFP started in auto but tripped on hydro coupling bearing temperature protection.

Action Taken:

- ✓ Modification was carried out using Instamatic B card VU-11, VO-11 & FA-12 cards. It is suspected that modification of VU-11, VO-11.
- ✓ The above cards kept under continuous monitoring.

Recommendations:

- Trip oil pressure low protection logic was changed from 1 out of 2 to 2 out of 3 logic.
- The above cards kept under continuous monitoring.

Case Study – 19.

Name of the Power Plant: FEROZE GANDHI UNCHAHAR THERMAL POWER PLANT, UNCHAHAR, RAEBARELI, UTTAR PRADESH.

Unit Running Status:

Load: 220MW

Date: 02.12.09

Name of the Equipment Failed: BOILER TRIPPED.

Cause: SLLD fully closed.

Observations:

- Unit was running at 220MW in EHC mode with hydraulic mode tracking EHC.
- Fuel master came down drastically and boiler tripped at 'Furnace pressure Very Low Protection'.
- Starting device started decreasing and went on to full closed position beyond the limiting value of 60%.
- Control valves got closed leading to increase in pressure causing reduction in fuel master.

Root Cause of failure:

- ❖ Mal-operation of driver card of Starting Device.
- ❖ SLLD fully closed.
- ❖ Control valves fully closed

Action Taken:

- ✓ Drive card of starting device was replaced.

Recommendations:

- Regular checking of drive card to be carried out.
- Checking of SLLD functioning at regular basis.

Case Study – 20.

Name of the Power Plant: FEROZE GANDHI UNCHAHAR THERMAL POWER PLANT, UNCHAHAR, RAEBARELI, UTTAR PRADESH.

Unit Running Status:

Load: 200MW

Date: 07.06.09

Name of the Equipment Failed: UNIT TRIPPED ON FLAME FAILURE.

Cause: PA header pressure went up.

Observations:

- Unit was running with mills A, C, D&E. Feeder A plugged alarm appeared and Mill A tripped on ignition permit not available.
- PA header pressure went up from 740 to 850mmwcl. Mill B was started.
- PA header pressure low alarm came and mills BCD tripped.
- Unit tripped on flame failure.

- When mill A tripped PA demand got reduced but PA fan A blade pitch did not operate as its actuator became inoperative. PA fan B got stalled as its blade pitch came down and PA pressure went low.

Root Cause of failure:

- ❖ PA header pressure went up from 740 to 850 mmwcl.
- ❖ PA pressure went low.

Action Taken:

- ✓ Monitoring of header pressure to be carried out before taking any mill in service.
- ✓ Maintaining header pressure between 700 to 800 mmwcl.

Recommendations:

- Regular checking of header pressure to be carried out.
- Provision online monitoring of header pressure.

Case Study – 21.

Name of the Power Plant: FARAKKA SUPER THERMAL POWER STATION, NTPC COAL STATION, FARAKKA (W.B)

Unit Running Status:

Load: 250MW

Date: 11.03.2010

Name of the Equipment Failed: UNIT TRIPPED ON FLAME FAILURE.

Cause: PA fan damper received a close command.

Observations:

- Unit tripped on flame failure during furnace disturbance following PA fan A tripping.
- The exact cause for PA fan tripping could not be identified.
- The Unit tripped again on 12.03.2010 & 15.03.2010 due to the same reason.
- After tripping on 15.03.2010, PA fan A outlet dampers' operation has been inhibited by isolating instrument air.

Root Cause of failure:

- ❖ The PA fan damper received a close command and its limit switches sensed “not open” thereby tripping the fan.
- ❖ The source/cause for this command could not be traced.

Action Taken:

- ✓ Site C&I and operation to observe closely for such spurious signal, if any, to pin point the root cause.

Recommendations:

- The reason for PA fan tripping from discharge damper “not open” to be investigated during the forthcoming overhaul and discharge damper operation to be normalized.

Case Study – 22.

Name of the Power Plant: KAHALGAON SUPER THERMAL POWER PLANT (KHSTPP),
KAHALGAON, BHAGALPUR, BIHAR

Unit Running Status:

Load: 0 MW

Date: 03.02.2010

Name of the Equipment Failed: UNIT TRIPPED.

Cause: Drum safety valves lifted at a lower pressure.

Observations:

- Unit tripped when three Drum safety valves lifted at a lower pressure due to an impulse, generated by bursting of pressure gauge cum switches' bourdon tube.
- Two valves could be reset by command but the third valve's plunger got stuck up.
- Unit tripped on Drum Level Low protection.

Root Cause of failure:

- ❖ Lifting of Drum Safety valves and subsequent non-resetting of one valve.

Action Taken:

- ✓ Preventive mtc. of solenoid operated all safety valves' solenoid and plunger to be carried out.
- ✓ Preventive mtc. of solenoid operated all safety valves' pressure switch to be carried out as per schedule.
- ✓ Pressure gauge cum switch to be replaced in all stage I units once every 5{five} years or earlier.

Recommendations:

- Two outages have taken place in the same unit within a span of few days due to failure / leakage of field device of Drum safety Valve. Routine walk down checks shall be done to identify such leakages beforehand and avoid such tripping.

Case Study – 23.

Name of the Power Plant: SIMHADRI SUPER THERMAL POWER PLANT, Visakhapatnam, Andhra Pradesh

Unit Running Status:

Load: 220MW

Date: 03.08.2009

Name of the Equipment Failed: UNIT TRIP.

Cause: Failure of RTD used for sensing PA fan A motor DE bearing temperature.

Observations:

- Nature of failure of RTD is not clear from trip analysis report.
- Responsibility for implementing recommendations, action plan and target date to be indicated in the trip analysis report.

Root Cause of failure:

- ❖ Failure of RTD used for sensing PA fan A motor DE bearing temperature.

Action Taken:

- ✓ Possibility of blocking of protection “PA & FD fans Tripping on bearing temperature high;” when there is an abnormal rate of rise of temperature as in case of RTD failure; would be explored during forth coming overhaul in unit-2.
- ✓ It is proposed to replace all protection circuit RTDs of all FD and PA Fans once in four years in both units as a preventive measure. Schedule to be reviewed periodically.
- ✓ Calibration of PA HDR pressure switches used for protection at regular intervals of 12 months.

Recommendations:

- As recorded in the trip report, PA header pressure dropped from 820 mmwc to 686 mmwc in 5 seconds. At 03:16:57 hrs PA header very low switches acted prematurely (Actual pressure was 686 mmwc but set point is 510 mmwcl). However based on the data available with us no conclusion could be made.

- Details of the PA header pressure switches calibration after the incident are also not available. However the required calibration need to be done and the root cause of the deviation between actual pressure and the set pressure at which the PA header pressure V.Low switch acted needs to be rectified.
- Presently, conventional RTD's are used for measuring/ protection of PA fan/ motor bearing temperatures at Simhadri. To enhance the reliability of this protection, it is proposed to consider the replacement of conventional RTD's by 46 headless RTD's. The headless RTD's have already been provided at RSTPS stage- II &III.

Case Study – 24.

Name of the Power Plant: FEROZE GANDHI UNCHAHAR THERMAL POWER PLANT, UNCHAHAR, RAEBARELI, UTTAR PRADESH.

Unit Running Status:

Load: 200MW

Date: 10.09.2009

Name of the Equipment Failed: UNIT TRIP.

Cause: Flame Failure.

Observations:

- Unit tripped on flame failure protection. Coal quality at the time of tripping was poor.

Root Cause of failure:

- ❖ Coal quality at the time of tripping was poor.

Action Taken:

- ✓ Blending of coal must be resorted to, to overcome the problems created by the low VM coal. To facilitate this, different quality coal storage must be done at identified locations.
- ✓ If feeding of moist coal is reported, Mill outlet temperature must be maintained to dry out the coal and kept under close observation to detect choking.
- ✓ Oil support should be taken to stabilize the flame during poor coal quality regimes.
- ✓ Since the flame in the furnace during poor coal quality regime is not very healthy, any operations likely to have an effect on the combustion (for example change in air flow) must be carried out gradually and with utmost care.

Recommendations:

- Action Plan to reduce trips on account of Flame Failure' to be provided and ensure its implementation.

Case Study – 25.

Name of the Power Plant: KORBA SUPER THERMAL POWER PLANT, KORBA,
CHHATISGARH

Unit Running Status:

Load: 250MW

Date: 13.09.2009

Name of the Equipment Failed: UNIT TRIPPED.

Cause: Tripped on flame failure and electrical disturbance.

Observations:

- Multi-unit tripping due to transmission line disturbance. One of the conductors of 400 kV Korba— Bhatapara line (Bus-4), Y-phase jumper, from the jack bus to the Wave Trap snapped from the straight conductor portion between the turn buckle and PG clamp and swung down to cause a fault with the 400kV R-phase Bus-3. Unit 6 tripped on bus bar protection. After a few minutes, Korba-Bhilai line 1 tripped due to phase to ground fault.
- This led to further tripping of Korba-Vindhyachal line 1 due to power swing and subsequent islanding of the station. Units 3, 5 tripped on over speed, unit 2 was manually tripped to prevent over speeding and unit 4 tripped on flame failure.
- Overloading of transmission lines to be avoided during such emergencies. The Korba – Bhilai line 1 was overloaded for 15 minutes before it tripped on phase to ground fault.
- The blackout could have been avoided by timely reduction of load in coordination with SCE.

Root Cause of failure:

- ❖ Transmission line disturbance a few minutes after tripping of unit 6. During power flow adjustments, Units 3, 5 tripped on over speed. Unit 2 was hand tripped to prevent over speed. Unit 4 tripped on flame failure.

Action Taken:

- ✓ Korba- Vindhyachal line-2 line failure of bus bar protection trip relay to trip the breaker is to be investigated and rectified.

- ✓ K-B 2 spurious command leading to tripping to be resolved with PGCIL.
- ✓ Failure of auto changeover in CW St-II to be checked and rectified.
- ✓ Failure of A/R in Korba-Bhatapara line to be attended.
- ✓ Failure of A/R in Korba-Bhilai 1 line to be attended.

Recommendations:

- Thermovision scanning of the switchyard needs to be done on a monthly basis.

Case Study – 26.

Name of the Power Plant: TALCHER THERMAL POWER STATION, TALCHER, ANGUL, ODISHA.

Unit Running Status:

Load: 250MW

Date: 31.08.2009

Name of the Equipment Failed: SWITCHYARD UNBALANCE.

Cause: Isolator jumper.

Observations:

- Ramchandrapur-Joda line was recharged following a fault tripping caused by snapping of one isolator jumper at joda substation. This led to a heavy voltage dip and unit tripped on negative phase sequence due to 132 KV system unbalance.

Root Cause of failure:

- ❖ 132KV system unbalance during a grid disturbance.

Action Taken:

- ✓ Reason for grid disturbance is to be investigated and the matter is to be taken up with SLDC to avoid the recurrence of such failures in future.

Recommendations:

- Excitation change over from auto to manual should be avoided.
- Voltage drop in 220/132KV System need to be promptly taken-up with SLDC for appropriate action to their end.

Case Study – 27.

Name of the Power Plant: FARAKKA SUPER THERMAL POWER STATION, NTPC COAL STATION, FARAKKA (W.B)

Unit Running Status:

Load: 250MW

Date: 07.05.2009

Name of the Equipment Failed: UNIT TRIP.

Cause: booster pump suction filter choking.

Observations:

- Booster pump suction filter choking.
- Drum level disturbance.
- Boiler trip due to level disturbance.
- Turbine trip due to boiler trip.
- Generator circuit breaker open due to turbine trip.

Root Cause of failure:

- ❖ BFP-B booster pump suction filter choking.

Action Taken:

- ✓ Before charging HP Heaters after maintenance work, proper flushing to be done and to be put in circuit after clearance from chemistry.
- ✓ Alarm analysis system in DDCMIS - to be commissioned.
- ✓ Consequences of Alarm appeared in LVS is to be analyzed and to be communicated accordingly.

Recommendations:

- Reason for BFP suction strainer choking within 11 days after overhauling may be analysed in detail and corrective action to be taken.
- Standby BFP starting permissive (Bearing temperature normal) is not available as alarm was persisting. This was not noticed by operation staff even after 10 hrs of appearing that alarm.
- This is a serious lapse on the part of operation. Any alarm concerned with the health of running/standby equipments shall be taken note of immediately and corrective action to be taken accordingly.

Case Study – 28.

Name of the Power Plant: KAHALGAON SUPER THERMAL POWER PLANT (KHSTPP),
KAHALGAON, BHAGALPUR, BIHAR

Unit Running Status:

Load: 250MW

Date: 04.09.08

Name of the Equipment Failed: UNIT TRIP.

Cause: Drum level Lo Lo.

Observations:

- Unit tripped on Drum Level Lo Lo when HP bypass fast open acted due to a lower set pressure. MS pressure, flow & FW flow reduced.
- The situation aggravated when TDBFP recirculation valves opened due to low flow.

Root Cause of failure:

- ❖ Opening of HP bypass valves caused by HP fast opening protection.

Action Taken:

- ✓ HP Bypass pressure set point put in auto with 10bar difference.

Recommendations:

- MS pressure deviation control set point to be monitored stringently and regularly.
- The possibility of HP bypass set point tracking even in manual mode needs to be explored.

Case Study – 29.

Name of the Power Plant: KAHALGAON SUPER THERMAL POWER PLANT (KHSTPP),
KAHALGAON, BHAGALPUR, BIHAR

Unit Running Status:

Load: 367 MW

Date: 06.10.2009

Name of the Equipment Failed: TG BEARING NOS. 1 & 3 GOT DAMAGED.

Cause: No lubricating oil supply to these bearings.

Observations:

- Unit was operating with seven mills at 367 MW with a coal flow of 380 MT/hr.
- Soot blowing operation at this load created furnace disturbance due to the following reasons.
 1. Pressure setting of the soot blower was on the higher side.
 2. Coal quality was poor.DC supply also failed to take over.
- From the trip report it is not clear whether the soot blowing operation was done in simplex mode or duplex mode. Pressure setting of the value of the lower elevation blowers is also not indicated.

Root Cause of failure:

- ❖ Flame failure.

Action Taken:

- ✓ Soot blowing operation reduced after higher furnace operations.

Recommendations:

- In addition to scheduled check in , in case of intermittent ‘no flame ‘ voting pressure setting of soot blowers (WW and LRSB) to be done.
- Soot blowing may be carried out in simplex mode during low load conditions.

Case Study – 30.

Name of the Power Plant: TALCHER SUPER THERMAL POWER STATION, ANGUL, ODISHA

Unit Running Status:

Load: 250MW

Date: 10.07.2009

Name of the Equipment Failed: UNIT TRIP.

Cause: High MS pressure.

Observations:

- Coal flow had been increased to increase the unit load. But load could increase only after increasing the load limiter set point.
- During load increase, MS pressure increased and ‘MS pressure Hi’ alarm was mistaken as ‘HPBP oil pressure Lo Lo’.
- HP bypass pressure went high and unit tripped on MS pressure before HPBP > 195 bar’ protection.
- HP bypass did not open as its set point was kept 197 Ksc by mistake.

Root Cause of failure:

- ❖ Insufficient operational checks during normal running leading to emergency during a routine load variation.

Action Taken:

- ✓ Mill put in auto.
- ✓ HP Bypass put in auto with 10 Bas plus pressure.

Recommendations:

- Max Load limiter set point should be kept at 530 MW and should not be disturbed in normal operation.
- HPBP oil pressure lo lo and MS pressure high alarm should be in separate facia window.
- Whenever MS pressure is increasing and HPBP valves are not opening on auto.
- MS pressure should be controlled with normal operation practices.

Case Study – 31.

Name of the Power Plant: FEROZE GANDHI UNCHAHAR THERMAL POWER PLANT, UNCHAHAR, RAEBARELI, UTTAR PRADESH.

Unit Running Status:

Load: 250MW

Date: 24.06.2009

Name of the Equipment Failed: UNIT TRIPPED ON “LOSS OF ALL FUEL”.

Cause: BFP scoop control failure.

Observations:

- Unit was running at full load. Disturbance in BFP Scoop/ feed water flow control occurred. While controlling unit load to handle the situation unit tripped on “loss of all fuel”.
- 1. Reason for reduction in PA header pressure is not clear from the analysis.
- 2. Reason for BFP scoop failure is not brought out in the analysis.
- 3. Remedial measures taken to rectify the BFP scoop control problem are also not

appearing in the report.

Root Cause of failure:

- ❖ BFP scoop control failure (Inadequate handling of emergency).

Action Taken:

- ✓ PA Header Pressure controller tuning to be checked.
- ✓ Purge Air flow for all Mills may be optimized by varying the damper openings so as to cause minimum furnace disturbance.

Recommendations:

- During purging of mill, withdrawing /cut in of oil guns should be avoided.
- Hot PA header pressure ‘actual’ variation should be in line of Transmitters raw values variation

Case Study – 32.

Name of the Power Plant: FEROZE GANDHI UNCHAHAR THERMAL POWER PLANT, UNCHAHAR, RAEBARELI, UTTAR PRADESH.

Unit Running Status:

Load: 250MW

Date: 14.02.2010

Name of the Equipment Failed: FURNACE PRESSURE LOW.

Cause: Furnace disturbance caused by PA fan flow disturbance.

Observations:

- While normalizing APH B after maintenance, running PA fan flows went to maximum and minimum respectively. PA header pressure came down and unit tripped on furnace pressure Lo Lo.

Root Cause of failure:

- ❖ Furnace disturbance caused by PA fan flow disturbance..

Action Taken:

- ✓ First APH B was started and then PA fan gates were open keeping “0” velocity.

Recommendations:

- It is better to keep only one PA fan in service whenever there is reduced firing through less than 04 mills.
- As per the report, PA fan stalling occurred while opening the gas outlet damper of APH B. This is not self-explanatory as PA fan stalling can occur during a change in the PA flow only. This may be analyzed further.

Case Study – 33.

Name of the Power Plant: VINDHYACHAL THERMAL POWER STATION, SINGRAULI, MADHYA PRADESH.

Unit Running Status:

Load: 250MW

Date: 09.09.2010

Name of the Equipment Failed: GENERATOR OVER-VOLTAGE.

Cause: Generator Lock out relay operated, Master fuel Trip, Turbine Over speed protection.

Observations:

- Unit was running at full load. At 17:27 Hrs. Unit tripped on “Generator Over-Voltage (STG-II)”.

- On investigation, Generator over voltage relay “59G2 - RXEG21 ABB make” was found operated.
- Relay got energized without generator terminal over voltage and remained energized even after unit tripping (i.e. no PT voltage condition).
- Defective 59G2 relay replaced.

Root Cause of failure:

- ❖ Generator over voltage protection relay was found defective.

Action Taken:

- ✓ D/A level controller VG43 became inoperative electrically due to Mechanical Jamming, the same was opened manually and D/A level was controlled.
- ✓ HP Bypass BP-1 servo valve found in disturbed condition, the same was replaced.

Recommendations:

- Relay testing is being done every overhaul by applying set voltage. Test voltage at threshold level to be kept at relay terminal for minimum of 30 minutes. Provision for pad locked for important isolators to avoid any accidental operation

Case Study – 34.

Name of the Power Plant: VINDHYACHAL THERMAL POWER STATION,
SINGRAULI, MADHYA PRADESH.

Unit Running Status:

Load: 500MW

Date: 29.10.2010

Name of the Equipment Failed: BOILER TRIP.

Cause: Water Wall circulation not correct.

Observations:

- MFT Operated.
- Turbine trip gear operated.
- Generator protection operated.
- BCW pp B parameters and other valve status got bad in MMI and at 12:00 hrs boiler tripped on Water Wall Circulation low.
- On analysis it was found that one card (drive control module-AS04) for station to U # 8 PRDS interconnection IBV in CRB09 panel (where BCWP B and other parameters are

housed) has malfunctioned causing failure of communication bus and erratic behavior of other modules of the panel which resulted closer of economizer inlet valve (E2) and feed water pump in MMI became fully inoperative, resulting falling of the FW flow to economizer.

Root Cause of failure:

- ❖ Drive control module (AS04) for IBV of IBV of U # 8 interconnecting PRDS vv malfunctioned.

Action Taken:

- ✓ Drive Control Module changed.

Recommendations:

- Replacement of faulty card (AS04) was done. Feedback taken from Dadri regarding similar problem and action taken there off at Dadari.
- Power supply of E2 v/v (FW eco inlet) is kept off to avoid such problem. Based on this feedback power of this v/v is kept off in all the units.

Case Study – 35.

Name of the Power Plant: VINDHYACHAL THERMAL POWER STATION,
SINGRAULI, MADHYA PRADESH.

Unit Running Status:

Load: 380MW

Date: 24.11.2010

Name of the Equipment Failed: UNIT TRIP.

Cause: Generator Low Forward Power Trip.

Observations:

- MFT operated.
- Turbine trip gear operated.
- Generator protection operated.
- Unit was running normal at 380 MW load and Unit tripped on 24.11.2010 at 05:54hrs on Gen low forward power protection.
- Subsequently turbine tripped and Boiler tripped on RH protection.
- Prior to tripping while raising load following observation was made:-
- Pressure dip observed in LP pr. of control fluid pp 2 which was in service.

- There was simultaneously dip in EHC demand leading to discrepancy and closing of control valves.

Root Cause of failure:

- ❖ Sluggishness in EHC Pilot Mechanical component.

Action Taken:

- ✓ Sludge removed from the line.
- ✓ Filter changed.

Recommendations:

- Regular checking of filter to be carried out.
- Differential Pressure to be measured around the filters.

Case Study – 36.

Name of the Power Plant: VINDHYACHAL THERMAL POWER STATION,
SINGRAULI, MADHYA PRADESH.

Unit Running Status:

Load: 510MW

Date: 20.12.2010

Name of the Equipment Failed: UNIT TRIP.

Cause: Loss of excitation.

Observations:

- Unit was on bar at full load. At 13:48:41 Hrs. “loss of excitation” protection operated and generator tripped. Thereafter turbine tripped followed by boiler tripping.
- After checking it was identified by EMD that PMG output AC 3 phase cable got earthed and found to be in burnt condition.
- Un-insulated bend was just touching the cable. The drip line is laid just above the Cable tray.

Root Cause of failure:

- ❖ Heating of PMG output 3 phase AC cable by HPH 5B un-insulated (1M length just below 17 M floor) alternate drip line.

Action Taken:

- ✓ Cable earthing point pin pointed and cable jointing done.
- ✓ Insulation provided over the un-insulated line.

Recommendations:

- HPH 5B alternate drip line.
- Identifying and rerouting of cables at vulnerable locations which may lead to cable heating.

Case Study – 37.

Name of the Power Plant: SIPAT SUPER THERMAL POWER STATION, SIPAT, BILASPUR (Chhattisgarh)

Unit Running Status:

Load: 500 MW

Date: 20.08.09

Name of the Equipment Failed: BOILER TRIP THROUGH REHEAT PROTECTION

Cause: Delay in opening of HP-LP Bypass valves when turbine tripped.

Observations:

In one of the 500 MW sets problem of Boiler tripping on either of the following protections was reported whenever turbine tripped:

- Re-heater protection.
- Boiler load index < 30%

Analysis

- ❖ Whenever turbine trips and boiler troubles are not present, later should continue to operate through HP-LP bypass system. Under such conditions Reheater is to be protected against starvation. So above two protections are provided. RH protection has a time delay of 9 seconds where as BLI has 3 seconds only.
- ❖ Trends analysis confirmed that with turbine trip outs instant commands were issued by HP BP as well as LP BP controllers were issued by HP BP as well as LP BP controllers to go to full open states. HP BP valves followed the commands simultaneously and were opening as per command within 1 sec. of issue of commands. However, LP BP valves were taking more than 20 seconds to open.
- ❖ The system was thoroughly checked and dampers provided in spray water pressure impulse line to limiter of LP BP rack were found to be welded in wrong direction.

Explanation

- ✚ To protect condenser against damages caused by hot steam entering through LP BP system condensate spray system is provided. To ensure that sufficient spray water is available before steam is dumped in condenser, the spray water pressure at the downstream of water injection valves is sensed and communicated to limiter which acts as the permissive providing tool for LP BP stop and control valves. This signal is also used as LP BP tripping signal and acts on injection water pressure slide valve in the rack.
- ✚ The mechanism has sensing of spray water pressure at downstream of WI valves via damper. When LPBP system is not in use, no water will be available in downstream. To avoid undesirable thrust on the bellow of limiter, it is necessary to ensure positive filling of impulse line up to the bellow. This is achieved by optimizing orifices (1) & (2). Damper also acts as NRV and stops back flow of fill-water when LPBP is not in use.
- ✚ The dampers were installed in the reversed direction. This not only was resulting into draining of spray water pressure impulse line but also the limiter was providing permissive to LPBP control and stop valves opening due to injection water pressure ahead of the water injection valves (due to stoppage of water flow at damper on pressurization of impulse line with spray water at downstream of WI valves).
- ✚ This whole process of communication of spray water pressure (that, too of upstream header) to limiter was taking considerable time (> 21 seconds).

Action Taken

The dampers were cut and re-welded in the right direction. Various trials were taken. The time gap between issue of command by controller (Askania) and execution of valve strikers could be reduced of 1 second.

Thus, the problem of Boiler trip, on turbine tripping due to above protections could be resolved.

RECOMMENDATIONS:

- Commissioning of all systems & sub systems should be carried out, if required, by simulation.
- Checking of Alarm, Interlock, and protection of the system.

Case Study –38.

Name of the Power Plant: OP JINDAL SUPER THERMAL POWER STATION, TAMNAR, RAIGARH, CHATTISGARH.

Unit Running Status:

Load: 250MW

Date: 08.04.11

Name of the Equipment Failed: GENERATOR ELECTRICAL TRIP

Cause: Heavy Earth Fault in 33KV JPL-CHP line

Observations:

- 33KV JPL – CHP line tripped through distance protection relay.
- On further checking a heavy earth fault noted near project gate around 1.5 KV from 33KV Switch Yard.
- Line insulator got burst & high current resulted failure of 33 KV cable near DM plant.
- As a result it created heavy voltage dip in the system.
- The 33KV line zone protection relay operated and isolated the faulty zone within 470 msecs.
- Generator got trip on Rotor Earth Fault.

Root Cause of failure:

- ❖ Faulty line insulator.
- ❖ Line insulator got burst & high current passed through earthing.

Action Taken:

- ✓ Replaced new tested insulator and charged 33KV line.

Recommendations:

- ✚ Regular checking of availability of line insulator.

Case Study – 39.

Name of the Power Plant: VINDHYACHAL THERMAL POWER STATION, SINGRAULI, MADHYA PRADESH.

Unit Running Status:

Load: 208 MW

Date: 05.11.10

Name of the Equipment Failed: TURBINE TRIP.

Cause: “Turbine trip protection operated”

Observations:

- Unit was running at full load, suddenly, Unit tripped on Turbine trip protection however the turbine speed rose to 3220 RPM.
- No relay/flag found operated on GRP panel. Turbine trip protection was the first event recorded in SOE and consequently Boiler also tripped.
- The tripping may be due to sudden water droplets falling from Roof on GRP panel, due to increased leakage in PRDS drain and shortening the trip contact of relay.

Root Cause of failure:

- ❖ The tripping may be due to sudden water droplets falling from Roof on GRP panel which may led to relay contact short circuiting.

Action Take:

- Complete sealing of NOP roof was ensured, PRDS drain leakages were arrested.

Recommendations:

- ✓ All the stage – I units PRDS floor roof to be ensured leakage free and drainage system to be ensured clear.

Case Study – 40.

Name of the Power Plant: VINDHYACHAL THERMAL POWER STATION,
SINGRAULI, MADHYA PRADESH.

Unit Running Status:

Load: 350MW

Date: 24.11.10

Name of the Equipment Failed: TURBINE TRIP, MASTER FUEL TRIP AND GENERATOR TRIP.

Cause: Emergency Trip from the Desk

Observations:

- Unit was running normal at 500 MW load and due to low schedule load was reduced to 350 MW.
- All TG bearing vibrations including bearing pedestal and shaft were observed on rising trend so Unit tripped from emergency push button from desk.
- Subsequently Boiler tripped on RH protection.
- Prior to tripping following observations were made:-
 - i. Chilling cold ambient condition.

- ii. Turbine Gland steam pressure was fluctuating during load reduction process and condenser vacuum maintaining much better at ~ -0.935 ksc
 - iii. Maximum vibration reached in bearing pedestal LPT casing rear (X) more than 100 micron at the time of tripping.
- HP shaft vibration at the time of tripping reached 487 micron (X), which was on increasing trend since C/V was throttled due load reduction.
 - During costing down vibrations were also high (max. 493, 1X).
 - Turbine came to barring speed in 45 minutes and all TSI parameters were normal.
 - LPH & HPH Level and extraction pressure, hot well and D/A level, temperature, CW flow and condensate parameters, HPT casing T/B temperature, Lube oil temperature etc. checked and found to be in order.

Root Cause of failure:

- ✓ Seal steam header Pressure fluctuation and less evacuation of gland leak off steam through GSC and seal steam exhauster due to malfunctioning of exhauster NRV, resulting the seal steam header pressure wide variation, even in negative side.
- ✓ Inadequate evacuation of seal steam from LPT glands might have led to localized condensation due to chilled ambient condition leading to vibration.

Action Taken:

- i. GSC exhauster discharge NRV servicing done.
- ii. Leak off steam controller fine tuning done from C&I side.
- iii. Pressure variation in seal steam header reduced to normal range.

Recommendations:

- Servicing of GSC exhauster NRV in every O/H.
- Servicing or Replacement of Leak off steam controller.

Case Study – 41.

Name of the Power Plant: BADARPUR THERMAL POWER STATION, BADARPUR, NEW DELHI.

Unit Running Status:

Load: 200 MW

Date: 30.10. 2009

Name of the Equipment Failed: GENERATOR ELECTRICAL TRIP

Cause: “Generator Over-Voltage (STG-II)”

Observations:

- LPH 2 extraction valve was getting a continuous close command.
- A mismatch was observed between Card output and ladder logic output of the PLC.
- During checking of the PLC card and ladder program, PLC went into force mode and seal steam valve closed in the process and unit tripped on condenser vacuum low.
- With OEM help, problem was found in logic implementation between SCADA & PLC controllers. This Logic has been modified in all Stage-I units.

Root Cause of failure:

- ✓ Both CPU of PLC system went to force mode.

Action Taken:

- Faulty logic control in PLC was rectified for all units.

Recommendations:

- ✓ Only proven logic should be implemented.

Case Study – 42.

Name of the Power Plant: TANDA THERMAL POWER STATION, AMBEDKAR NAGAR, UTTAR PRADESH.

Unit Running Status:

Load: 200 MW

Date: 10.11.2009

Name of the Equipment Failed: BOILER TRIP

Cause: Unit flame failure

Observations:

- During coal flow variation caused by feeder A shear pin failure, boiler tripped on flame failure.
- Coal analyzed was having low VM and high Ash.

Root Cause of failure:

- ✓ Flame failure during coal variation.

Action Taken:

- Air flow & coal firing to be gradually adjusted, keeping control on Furnace parameters like O₂ , Furnace Draft etc.
- A number of Unit trippings while cut in / cut out of a Mill / Feeder.

Recommendations:

- While cut-in/ cut-out of Mills, air flow & coal firing to be gradually adjusted, keeping control on Furnace parameters like O₂ , Furnace Draft etc.
- In TANDA, there have been a number of Unit trippings while introducing/removing a Mill/Feeder.
- Operation Personnel should be better trained and detailed operating instructions need to be made available to the desk operators for carrying out various operations including Mill starting/stopping/purging etc.
- During Mill changeovers and other any other operation affecting furnace stability, a very close watch needs to be maintained on the flame scanners, particularly while low VM coal is being fired, and in case prolonged disturbance in flame is observed, oil support should be taken for stabilization.
- Secondary air dampers must be adjusted, particularly bottom most AA damper.

Case Study – 43.

Name of the Power Plant: TANDA THERMAL POWER STATION, AMBEDKAR NAGAR, UTTAR PRADESH.

Unit Running Status:

Load: 200 MW

Date: 18.11.2009

Name of the Equipment Failed: BOILER TRIP

Cause: coal flow variation

Observations:

- Unit tripped on furnace pressure Lo Lo.
- Prior to tripping, furnace was in disturbed condition and oil support was taken 5 minutes before tripping.
- A sudden loading/unloading of Mill A was observed just before tripping

Root Cause of failure:

- ✓ Furnace disturbance with coal flow variation

Action Taken:

- Mill A roller freeness checking during PM.
- Feeder A Dyna drive checking during P

Recommendations:

- ✓ In TANDA, there have been a number of Unit trippings while introducing /removing Mill/Feeder. The furnace gets disturbed with even slight variations in any of the operating parameters in the Boiler. To overcome this problem, auto operation of all the combustion related control loops needs to be introduced so that the requirement of manual intervention and the possibility of human errors can be minimized.
- ✓ Secondary air dampers must be accordingly, and immediately, adjusted after any Mill Change over.
- ✓ The reason behind the disturbance in the PA flow to Mill A needs to be investigated and attended. Detailed mapping of PA damper position and PA flow may be done to pin-point the defect.

Case Study – 44.

Name of the Power Plant: SINGRAULI SUPER THERMAL POWER PLANT,
SHAKTINAGAR, SONEBHADRA, UTTAR PRADESH

Unit Running Status:

Load: 200MW

Date: 12.05.2009

Name of the Equipment Failed: TURBINE TRIP

Cause: “Oil leakage in emergency governor line”.

Observations:

- Overlooking of maintenance of over speed trip device has resulted in unit tripping.

Root Cause of failure:

- Oil leakage in emergency governor line.

Action Taken:

- Front pedestal inspection window to be opened up to check for leakage during annual shutdowns.

Recommendations:

- During overhauling whenever governing system work is being carried out, DPT for all joints is required to be done to detect any development of crack.

- Work of assembly of governing system components should be done by experienced personnel under expert supervision to avoid improper assembly and subsequent damage to the union joint due to thread damage or pipe being under stress.
- Set of pipe and fittings of governing system be kept available to replace any component detected to have been even slightly damaged.
- Over speed feedback mechanism rubber roller and striker gap has to be checked and ensured.

Case Study – 45.

Name of the Power Plant: SIPAT SUPER THERMAL POWER STATION, SIPAT, BILASPUR (Chhattisgarh)

Unit Running Status:

Load: 500MW

Date: 02.05.2009

Name of the Equipment Failed: TURBINE TRIP

Cause: “Inadequate water circulation protection& MDBFP Booster pump decoupled”.

Observations:

- Unit tripped on inadequate water circulation protection.
- TDBFP –A and MDBFP were in service.
- MDBFP Booster pump decoupled.
- Though the load was reduced drum level could not be maintained as CRH was not lined up for TDBFP –A.

Root Cause of failure:

- MDBFP Booster pump decoupled.

Action Taken:

- CRH steam to be made available by lining up of valve.
- MDBFP Booster pump coupling failure to be taken up with M/S BHEL.

Recommendations:

- A circular has been issued on 0705/2009 to Operation shift personnel for avoiding bypassing of Drum level.
- Similar incident took place in U#4 on 11/4/09 where Drum level protection was bypassed.
- Bypassing of protections should be strictly avoided.

Chapter 4

CONCLUSION & RECOMMENDATIONS

CONCLUSION

Various power plants have developed technical expertise in different areas such as turbine, boilers, C&I etc to deal with emergencies inclusive of trippings analysis. Outcome of such exercises is documented and made available with concerned authorities for future linkage of needy power personnel (including learners, researchers and practicing professional) at a later stage to benefit the power industry at large. This work and subsequent report is based on various tripping of different TPSs, documents of which were available different authorities of JIPT and JPL, Tamnar, Raigarh.

The well-run plants have a trip committee, which is entrusted with the task of root cause analysis of trips and suggesting corrective actions to prevent recurrence of trips. The suggested corrective actions are typically formulated as an action plan with clearly ear-marked responsibility center and schedule. Compliance with such recommendations is monitored at plant as well as corporate levels. Recommendations of the trip committee also feed into the maintenance plan.

In some cases, specialized committees are also in place for analyzing boiler tube leakages – one of the most frequent reasons for forced outages. Other causes of forced outage are also analyzed in detail by what are called as ‘Forced Outage Committees’

While doing the report work we came across many tripping reasons, some of the main reasons of tripping of the unit are:

- Problem related with electrical power supply.
- Problem related with Fuel supply.
- Problem related with pressure drop in Lubrication oil line and in cooling water line.
- Problem related with supply failure related to feed water supply, steam supply etc.
- Human error with operating from the desk.

RECOMMENDATIONS:

These problems listed above can be reduced to minimum only by safe working and guiding our operators to proper utilization and proper handling of the desk board.

By using the result of R&D in this programme, plant owner and operators can:

- Reduce the cost of the loss of availability of the plant due to any tripping happen in the plant.
- Increase the safety and efficiency of the plant.
- This report can also help them to increase the availability of an auxiliary and life of the auxiliary.
- Conducting Regular Seminar on Thermal Power Plant Problems faced and its solutions done will help to reduce the hazards in Power Plants.

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